ML-Based Classification of Text Documents

1. Abstract

Text document classification is one of the most important applications of Machine Learning and Natural Language Processing (NLP). It involves automatically assigning documents to predefined categories based on their content.

This project develops a Machine Learning-based Text Document Classification System using the 20 Newsgroups Dataset. The system learns patterns from thousands of text documents and automatically predicts the category of a new document. The project demonstrates the complete machine learning workflow including data preprocessing, feature extraction, model training, evaluation, and prediction.

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2. Introduction

With the rapid growth of digital information, huge amounts of text data are generated daily through emails, news articles, blogs, and social media posts. Manually organizing and categorizing these documents is time-consuming and inefficient.

Machine Learning provides automated methods for analyzing textual information and classifying documents into different categories. Text classification is widely used in news categorization, email filtering, sentiment analysis, recommendation systems, and information retrieval.

This project focuses on building a machine learning model capable of automatically classifying text documents into their correct categories.

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3. Problem Statement

Organizations and users deal with large collections of textual documents. Manually sorting these documents into categories requires significant effort and time.

The goal of this project is to develop an automated text classification system that can accurately predict the category of a document based on its textual content.

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4. Objectives

The main objectives of this project are:

- To understand text classification techniques.
- To preprocess textual data using NLP methods.
- To convert text documents into numerical features.
- To train machine learning classification models.
- To evaluate model performance.
- To automatically classify documents into their correct categories.

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5. Dataset Description

Dataset Name: 20 Newsgroups Dataset

Source:
https://www.kaggle.com/datasets/crawford/20-newsgroups

Description:

The dataset contains approximately 20,000 text documents divided across 20 different newsgroup categories.

Number of Documents:

- Approximately 18,846 documents

Number of Categories:

- 20 Categories

Examples of Categories:

- comp.graphics
- comp.os.ms-windows.misc
- comp.sys.ibm.pc.hardware
- comp.sys.mac.hardware
- rec.autos
- rec.motorcycles
- rec.sport.baseball
- rec.sport.hockey
- sci.crypt

- sci.electronics
- sci.med
- sci.space
- talk.politics.misc
- talk.religion.misc

Dataset Characteristics:

Feature	Description
Text	Document Content
Target	Category Label

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6. Machine Learning Workflow

The project follows the standard machine learning process:

1. Data Collection
2. Data Exploration
3. Data Preprocessing
4. Feature Extraction
5. Dataset Splitting
6. Model Training
7. Model Evaluation
8. Prediction

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7. Exploratory Data Analysis (EDA)

Before training the model, the dataset is analyzed to understand:

- Number of documents
- Distribution of categories
- Average document length
- Most common words

Benefits of EDA:

- Understand dataset structure
- Detect imbalance
- Identify noisy data

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8. Data Preprocessing

Text data must be cleaned before being used for machine learning.

Preprocessing Steps

1. Lowercase Conversion

All text is converted into lowercase.

Example:

Before:

Machine Learning is AMAZING

After:

machine learning is amazing

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2. Removal of Punctuation

Special characters are removed.

Examples:

! ? @ # $ % ^ & *

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3. Tokenization

Documents are divided into individual words.

Example:

"Machine learning is powerful"

becomes

["machine", "learning", "is", "powerful"]

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4. Stop Word Removal

Frequently occurring words with little meaning are removed.

Examples:

- the
- is
- am
- are
- and
- of

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5. Stemming

Words are reduced to their root form.

Examples:

running → run

playing → play

studies → studi

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9. Feature Extraction

Machine Learning algorithms require numerical input.

Therefore, textual documents are converted into numerical vectors.

TF-IDF Vectorization

TF-IDF stands for:

Term Frequency – Inverse Document Frequency

It measures how important a word is within a document compared to the entire collection.

Advantages:

- Captures meaningful words
- Reduces effect of common words
- Improves classification accuracy

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10. Dataset Splitting

The dataset is divided into:

Dataset	Percentage
Training Data	80%
Testing Data	20%

Training Data:

Used to train the model.

Testing Data:

Used to evaluate performance.

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11. Machine Learning Algorithm

Multinomial Naive Bayes

The primary algorithm used for this project is Multinomial Naive Bayes.

Reasons:

- Excellent for text classification.
- Fast training and prediction.
- Works efficiently with TF-IDF features.
- Simple implementation.

Working Principle:

The algorithm calculates probabilities for each category and assigns the category with the highest probability.

12. Model Training

The model is trained using:

- Preprocessed documents
- TF-IDF vectors
- Training dataset

The model learns relationships between words and categories.

Examples:

Words like:

- graphics
- image
- computer

may indicate:

comp.graphics

Words like:

- medicine
- doctor
- disease

may indicate:

sci.med

Words like:

- hockey
- player
- team

may indicate:

rec.sport.hockey

13. Model Evaluation

The trained model is evaluated using testing data.

Evaluation Metrics

Accuracy

Measures overall correctness.

Accuracy = Correct Predictions / Total Predictions

Precision

Measures the quality of positive predictions.

Recall

Measures how many actual class instances were correctly identified.

F1 Score

Balances Precision and Recall.

Higher F1-score indicates better performance.

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14. Confusion Matrix

A confusion matrix is used to visualize classification performance.

It shows:

- Correct classifications
- Incorrect classifications
- Frequently confused categories

Benefits:

- Better understanding of model performance.
- Helps identify problematic categories.

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15. Results

After training and testing, the model successfully classified text documents into their respective categories.

Typical Results:

- Accuracy: 80% – 90%
- High Precision
- High Recall
- Good Generalization Performance

The model effectively learns category-specific vocabulary and predicts document classes accurately.

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16. Advantages

- Automatic document categorization.
- Fast classification.
- Handles large datasets.
- Reduces manual effort.
- Scalable for real-world applications.

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17. Limitations

- Performance depends on training data quality.
- Similar categories may cause confusion.
- Cannot fully understand document context.
- Accuracy may decrease on unseen topics.

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18. Applications

Text document classification is widely used in:

- News Categorization
- Email Classification
- Content Recommendation
- Search Engines
- Digital Libraries
- Customer Support Systems
- Social Media Monitoring

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19. Future Enhancements

Future improvements may include:

- Support Vector Machines (SVM)
- Random Forest Classifier
- Deep Learning Models
- LSTM Networks
- Transformer Models
- BERT-based Classification
- Real-time Document Classification System

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20. Conclusion

This project successfully demonstrates the use of Machine Learning and Natural Language Processing for automatic text document classification. The 20 Newsgroups Dataset was preprocessed and transformed into numerical features using TF-IDF vectorization. A Multinomial Naive Bayes classifier was trained and evaluated to categorize documents into one of twenty predefined categories. The model achieved strong classification performance and provided valuable insights into text mining and machine learning workflows. This project serves as an excellent introduction to text analytics and document classification systems.